

Dhruv Rath

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SUMMARY

Final-year B.Tech student building production-ready, scalable full-stack applications, LLM-powered systems, and intelligent RAG pipelines using JavaScript, Python, Node.js and SQL, with a strong focus on backend architecture, API-driven development, and AI & GenAI integration.

EDUCATION

Vellore Institute of Technology, Chennai

2023 – present (2027)

B.Tech - Electronics and Computer

CGPA - 9.06

Relevant Courses: Data Structures & Algorithm, OOP, Database Management, Operating Systems, Computer Networks, Computer Architecture, Cloud Computing, Web Programming, AI & ML, Deep Learning

EXPERIENCE

AI & ML Intern

June 2026 – August 2026

Sutherland

Chennai, Tamil Nadu

- Engineered backend services and data pipelines using Python and FastAPI, integrating vector databases and modular APIs to support scalable AI/LLM workflows.
- Developed Agentic AI workflows with modular tool integration and task orchestration for LLM-driven multi-step task execution and automation.

SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL

Frameworks & Libraries: Node.js, Express.js, FastAPI, LangChain, Streamlit, PyTorch, TensorFlow

AI: RAG, Agentic AI, LLMs, NLP, Machine Learning, Deep Learning, Prompt Engineering

Tools & Databases: MySQL, PostgreSQL, JWT Auth, Bcrypt, ChromaDB, Git, GitHub

PROJECTS

ShopEase – Full-Stack E-Commerce Platform [GitHub](#) | [Live](#) *Node.js, Express.js, MySQL, JavaScript, JWT*

- Architected a full-stack e-commerce system with Node.js, Express.js, MySQL, and RESTful APIs for catalog, cart, orders, users, and inventory.
- Engineered JWT authentication, bcrypt security, and RBAC with protected customer/admin endpoints.
- Integrated Razorpay payment processing with payment creation and server-side verification workflows.

MediDesk – Healthcare Appointment Manager [GitHub](#) | [Live](#) *Node.js, Express.js, PostgreSQL, React*

- Architected a concurrency-safe scheduling engine using Postgres advisory locks, preventing double-booking under concurrent requests.
- Engineered async pipelines for email delivery, Google Calendar sync and LLM-powered visit summaries.
- Implemented JWT-based RBAC across patient, doctor, and admin portals with encrypted OAuth tokens.

Hybrid-RAG Document Chatbot [GitHub](#) *Python, FastAPI, LangChain, JavaScript, ChromaDB*

- Built a modular full-stack RAG system for multi-format document Q&A and session management.
- Engineered a hybrid retrieval pipeline combining dense vector search, BM25, and cross-encoder reranking via FastAPI and LangChain for high-precision document retrieval.
- Designed session-isolated document pipelines for multi-user processing.

CERTIFICATIONS

- IBM – Advanced Generative AI - Professional
- Oracle – Agentic AI Certified Foundations Associate
- IBM – Prompt Engineering for Everyone
- Infosys Springboard – Machine Learning Implementation

ACHIEVEMENTS

- Shortlisted among 416 of 3,01,066 teams (**Top 0.14%**) in the Adobe University Hackathon 2026.
- Design Lead for the Sports Club at VIT Chennai